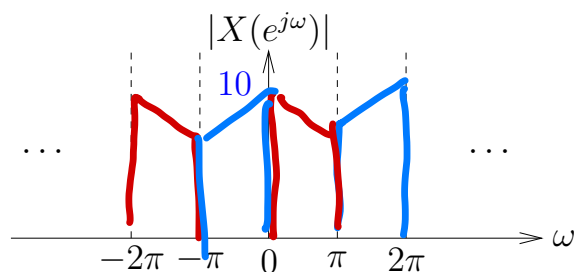
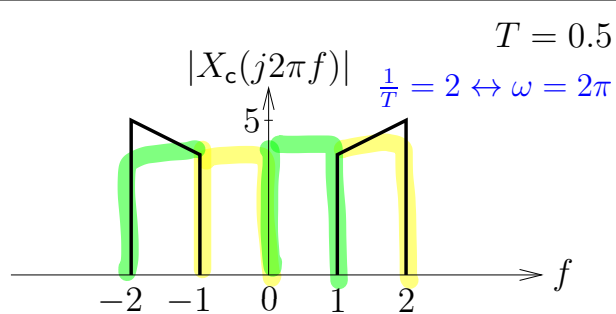
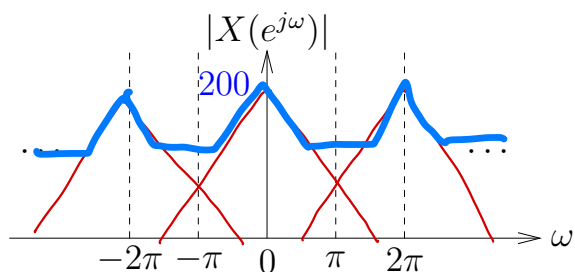
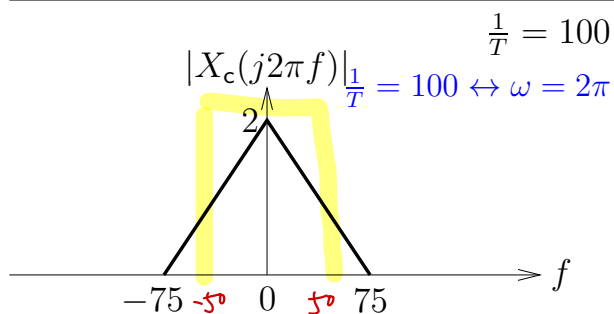
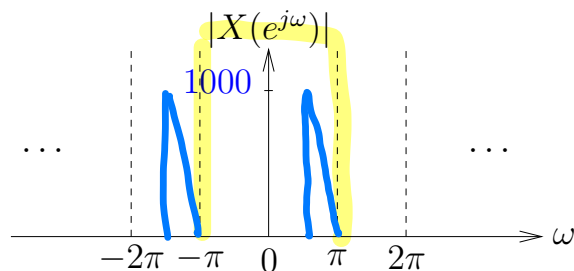
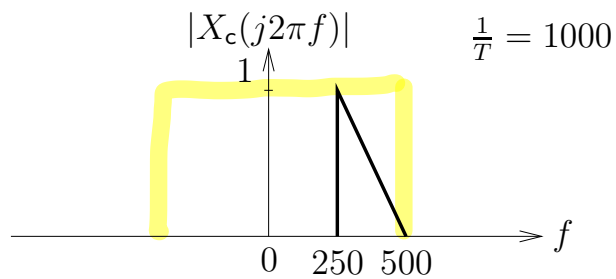
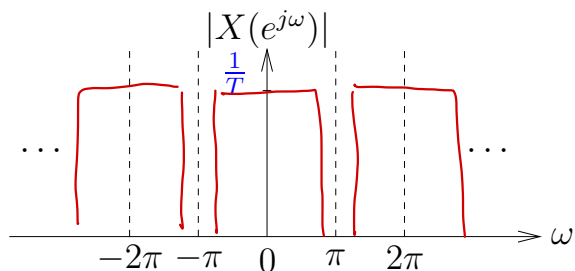
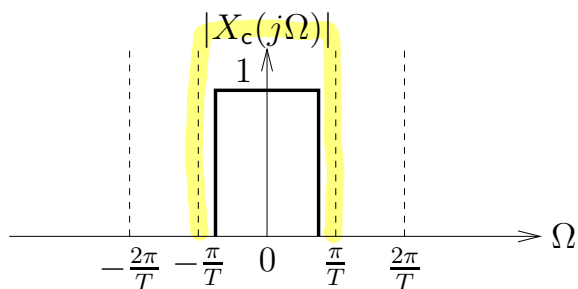


Sampling — Examples:

$$T\Omega = \omega$$

$$\Omega \in \left[-\frac{\pi}{T}, \frac{\pi}{T}\right]$$

$$f \in \left[-\frac{1}{2T}, \frac{1}{2T}\right]$$



Remember:

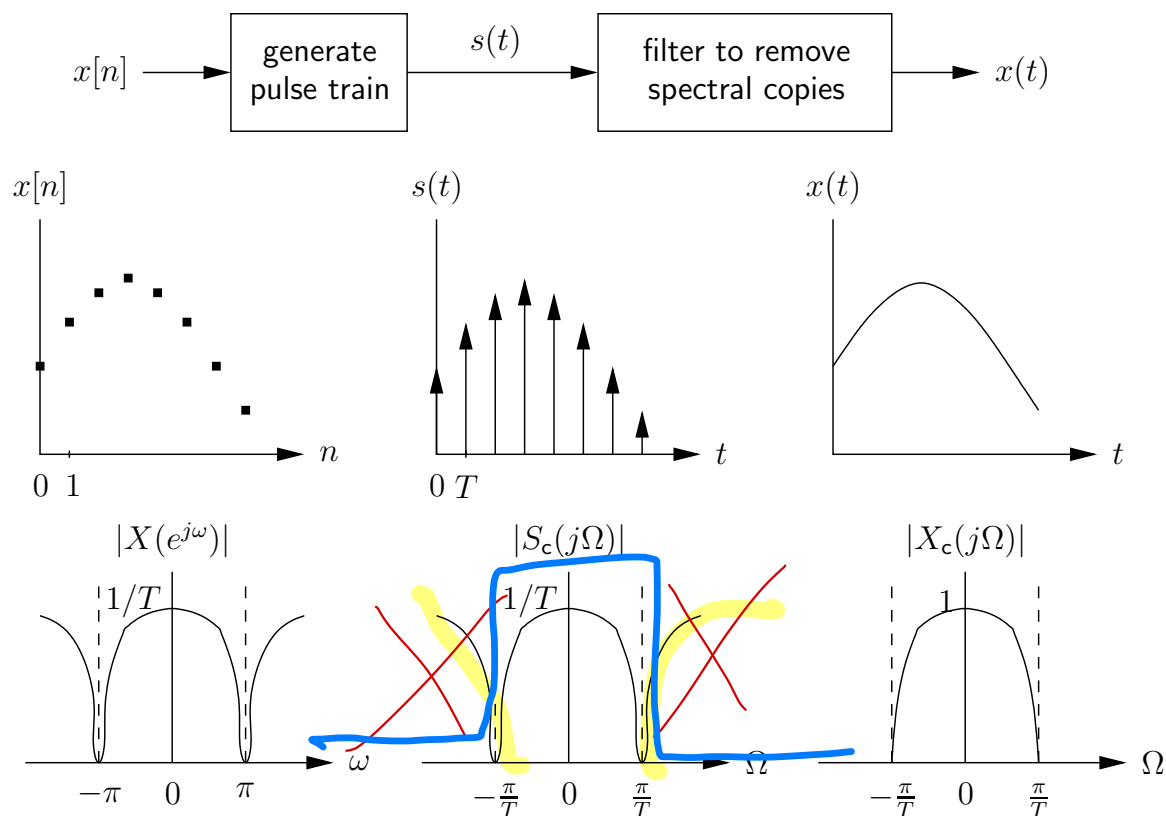
The sampling freq $\begin{cases} f = \frac{1}{T} \text{ Hz} \\ \Omega = \frac{2\pi}{T} \frac{\text{rad}}{\text{sec}} \end{cases}$ maps to $\omega = 2\pi \frac{\text{rad}}{\text{samp}}$.

Sinc Reconstruction:

- We will now detail the reconstruction of waveform $\{x(t)\}_{\forall t}$ from its samples $\{x[n]\}_{\forall n}$.
- Throughout, we'll assume that $x(t)$ is a lowpass signal (i.e., it lives in the 0^{th} Nyquist zone).
- For our CTFT/DTFT derivation, we viewed sampling as:

$$x(t) \rightarrow s(t) \rightarrow x[n].$$

We will now reverse this procedure for reconstruction.



Note: the frequency-domain plots suggest exactly the form of filtering that we need!

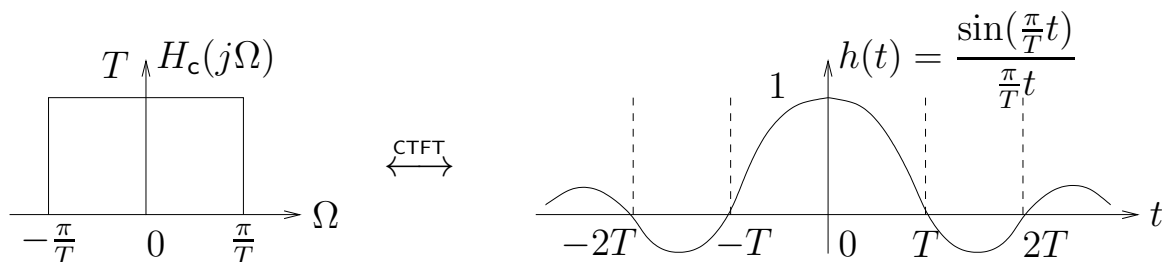
Sinc Reconstruction (cont.):

- To generate the pulse train $s(t)$, we can (in theory) do

$$s(t) = \sum_{n=-\infty}^{\infty} x[n]\delta(t - nT),$$

which places a $x[n]$ -scaled Dirac delta at all times $t = nT$.

- To obtain $x(t)$ from $s(t)$, the CTFT plots show that we need a “brickwall” lowpass filter with cutoff freq $\Omega_0 = \frac{\pi}{T}$ and gain T :



- Actually, we can combine these two operations into one:

$$\begin{aligned} x(t) &= \int_{-\infty}^{\infty} s(\tau)h(t - \tau)d\tau \\ &= \int_{-\infty}^{\infty} \sum_n x[n]\delta(\tau - nT) \frac{\sin(\frac{\pi}{T}(t - \tau))}{\frac{\pi}{T}(t - \tau)} d\tau \\ &= \sum_n x[n] \frac{\sin(\frac{\pi}{T}(t - nT))}{\frac{\pi}{T}(t - nT)} \end{aligned}$$

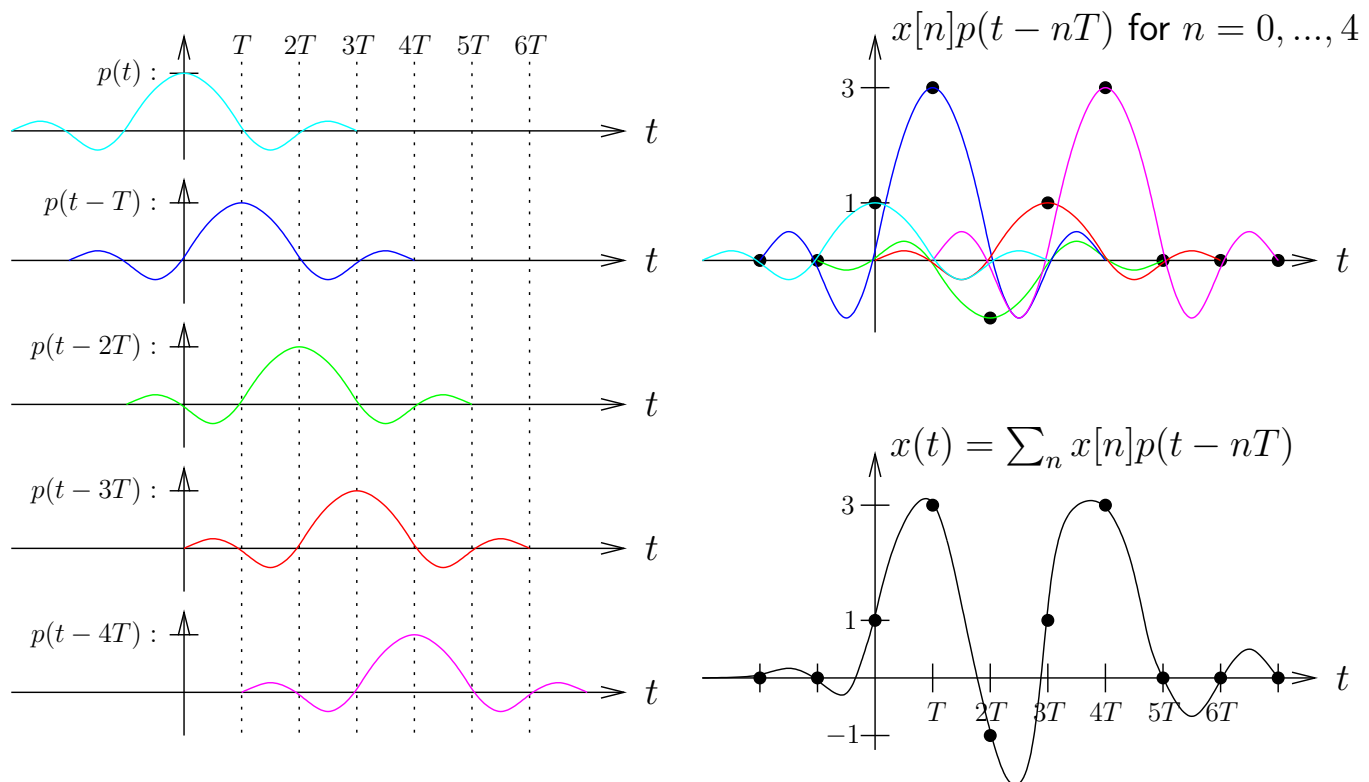
known as “sinc reconstruction.”

Sinc Reconstruction (cont.):

Sinc reconstruction:

$$x(t) = \sum_{n=-\infty}^{\infty} x[n] \frac{\sin\left(\frac{\pi}{T}(t - nT)\right)}{\frac{\pi}{T}(t - nT)}$$

interpolates the samples $\{x[n]\}$ using a sinc pulse “ $p(t)$ ” with T -spaced zero-crossings:



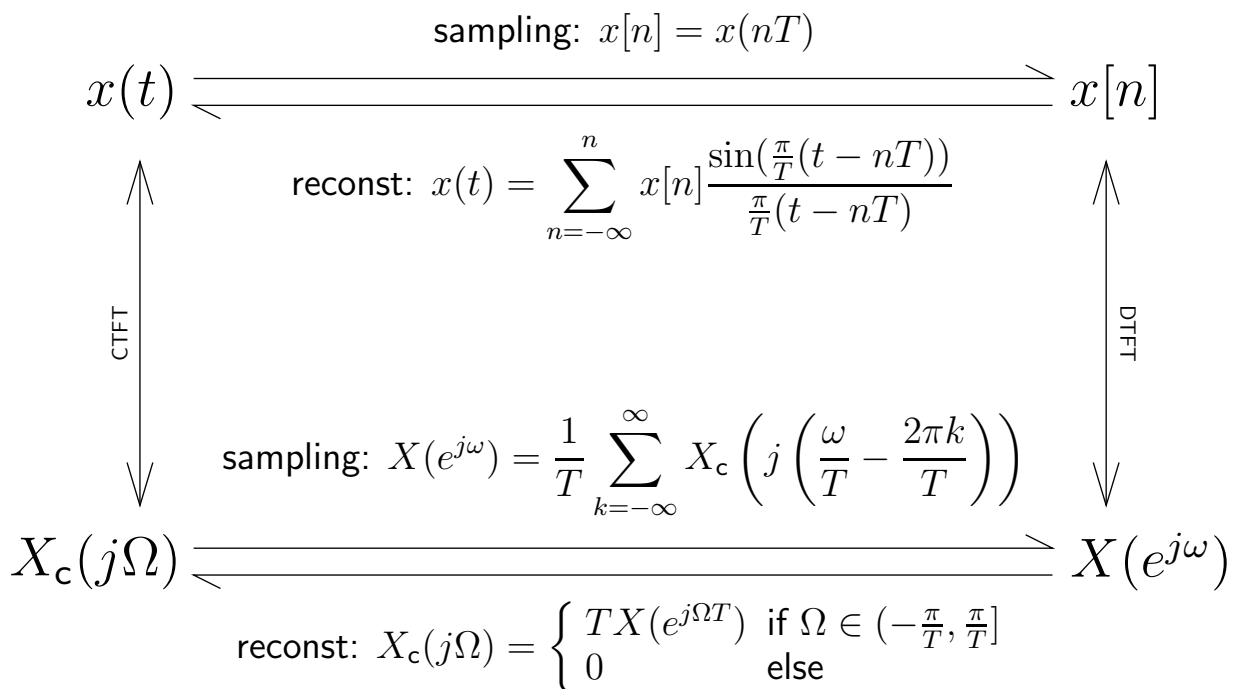
Note: the same procedure is called “Nyquist pulse shaping” in the context of digital communications.

Sampling and Reconstruction — Summary:

In the frequency domain, we saw that sinc reconstruction applies a LPF, with gain- T and cutoff $\frac{\pi}{T}$ rad/sec, to $S_c(j\Omega)$. We know $S_c(j\Omega) = X(e^{j\Omega T})$, and thus when $x(t)$ is bandlimited to $\frac{1}{2T}$ Hz, so that no aliasing occurs,

$$X_c(j\Omega) = \begin{cases} TX(e^{j\Omega T}) & \text{if } \Omega \in (-\frac{\pi}{T}, \frac{\pi}{T}] \\ 0 & \text{else.} \end{cases}$$

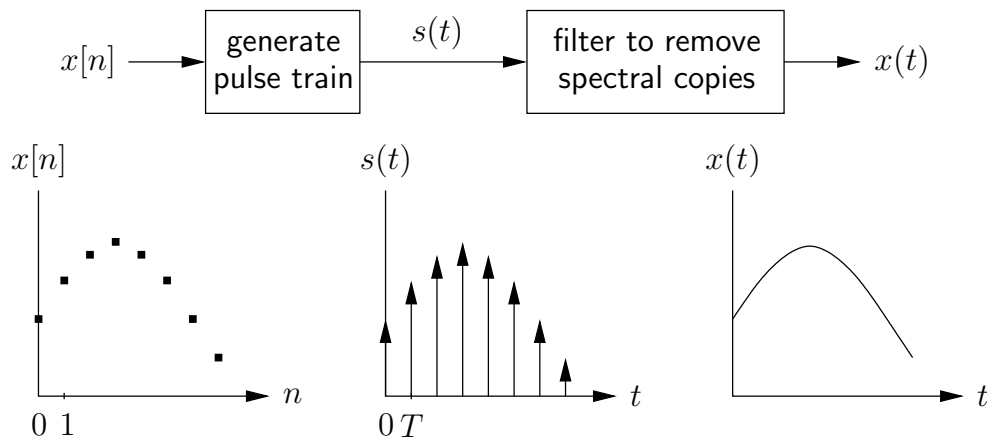
In summary, we have



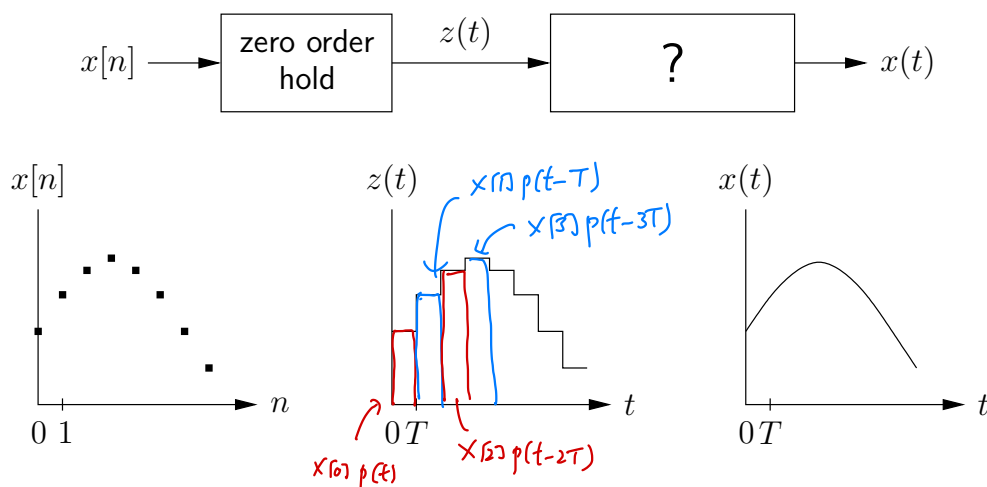
where the reconstruction equations hold *only* when $x(t)$ is bandlimited to $\frac{1}{2T}$ Hz.

ZOH Reconstruction:

- The “CT sampled signal” $s(t)$, a Dirac-delta pulse train, is impractical to generate. (It has infinite amplitude!)



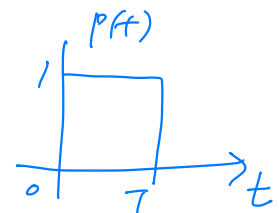
- Practical Idea: Use a rectangular pulse to generate a “zero order hold” signal $z(t)$ instead of a pulse train $s(t)$.



Is perfect reconstruction still possible? If so, how?

$$s(t) = \sum_n x[n] \delta(t-nT)$$

$$s(t) * p(t) = \sum_n x[n] p(t-nT)$$

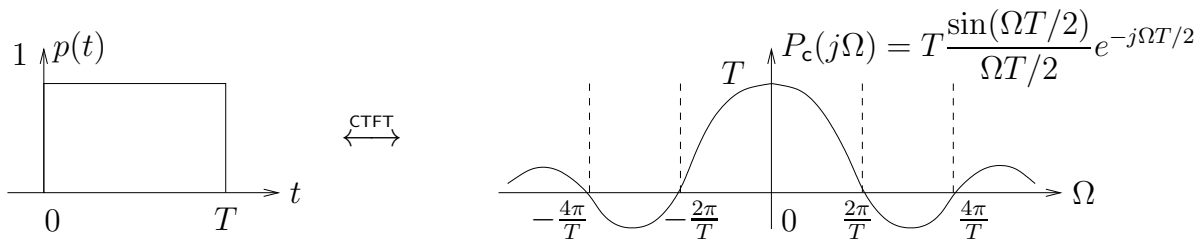


ZOH Reconstruction (cont.):

The ZOH signal $z(t)$ is related to the samples $x[n]$ via

$$z(t) = \sum_{n=-\infty}^{\infty} x[n]p(t - nT)$$

where $p(t)$ is a T -wide rectangular pulse:

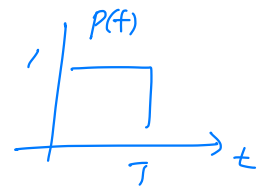


In the frequency domain, we find

$$\begin{aligned} Z_c(j\Omega) &= \int_{-\infty}^{\infty} \left(\sum_n x[n]p(t - nT) \right) e^{-j\Omega t} dt \\ &= \sum_n x[n] \int_{-\infty}^{\infty} p(t - nT) e^{-j\Omega t} dt \\ &= \sum_n x[n] e^{-j\Omega T n} P_c(j\Omega) \\ &= X(e^{j\Omega T}) \cdot T \frac{\sin(\Omega T/2)}{\Omega T/2} e^{-j\Omega T/2} \\ &= \frac{\sin(\Omega T/2)}{\Omega T/2} e^{-j\Omega T/2} \sum_{k=-\infty}^{\infty} X_c(j(\Omega - \frac{2\pi}{T}k)). \end{aligned}$$

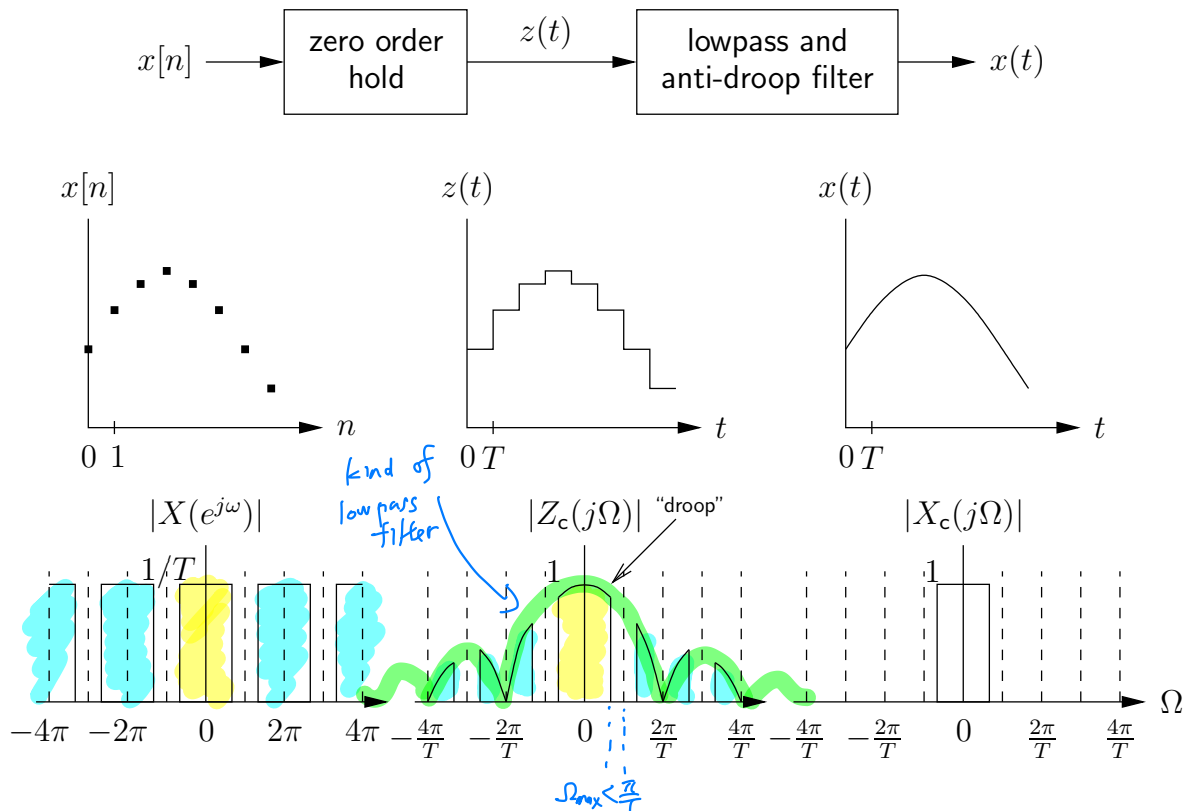
Comparing to $S_c(j\Omega)$, the effect of ZOH becomes clear!

$$z(t) = s(t) * p(t)$$

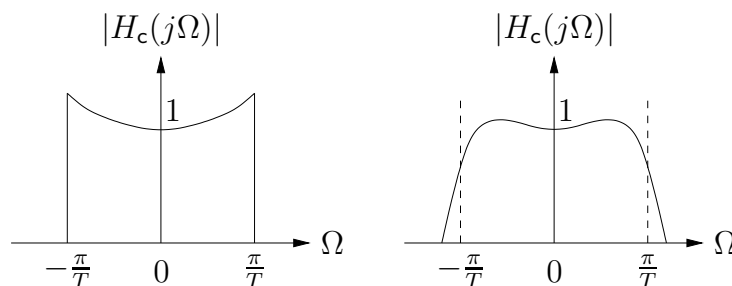


ZOH Reconstruction (cont.):

Summary of ZOH reconstruction:

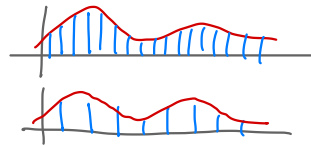


The 2nd stage of ZOH reconstruction is filtering that suppresses spectral copies and corrects for passband “droop”. Ideal (left) and practical (right) versions of this LPF look like:

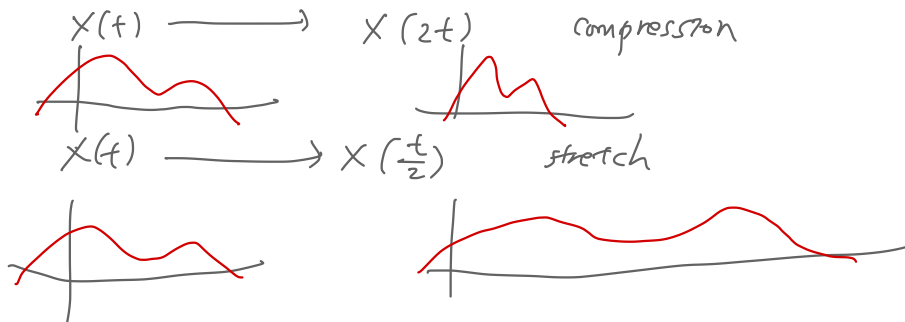


where the passband gain on the left is $\left| \frac{\sin(\Omega T/2)}{\Omega T/2} \right|^{-1}$.

Rate Conversion:

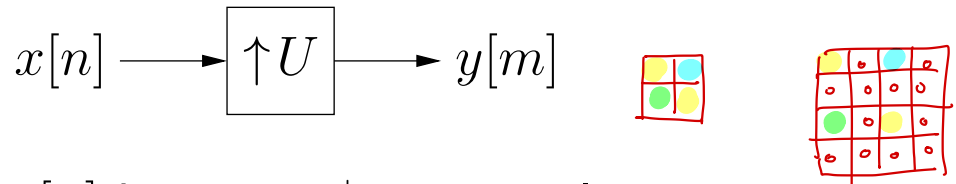


- Sometimes we would like to change the effective sampling rate of a signal *after* it has been sampled.
- To do this, we combine one of the elementary operations
 - *downsampling* (throwing away samples)
 - *upsampling* (inserting zero-valued samples)with discrete-time lowpass filtering in order to perform
 - *decimation* (decreasing effective sampling rate) and
 - *interpolation* (increasing effective sampling rate),respectively. These last two procedures change the effective sampling rate by an integer factor.
- When we want to change the effective sampling rate by a rational factor $\frac{U}{D}$, we combine interpolation and decimation into what is known as *resampling*.

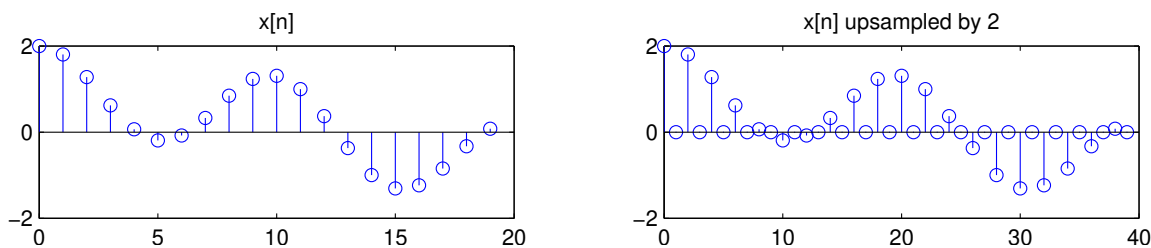


Upsampling:

- We represent upsampling by the block diagram



- To upsample $x[n]$ by $U \in \mathbb{Z}^+$, we simply insert $U - 1$ zeros between each sample in $x[n]$.

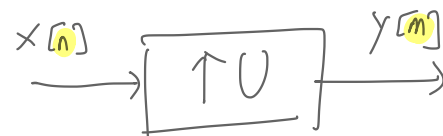


- In the time domain, upsampling can be written as

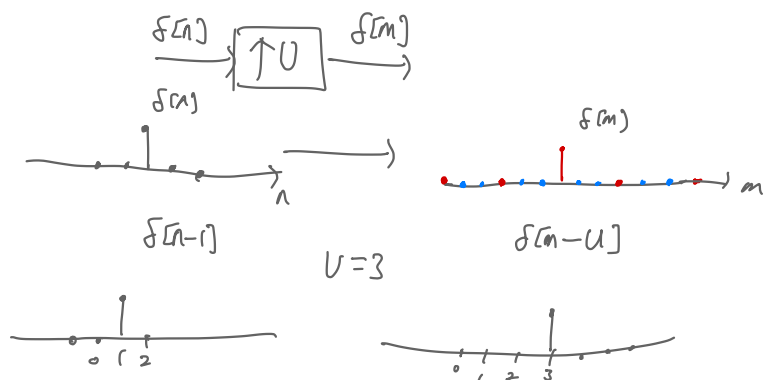
$$y[m] = \begin{cases} x[\frac{m}{U}] & \text{if } \frac{m}{U} \in \mathbb{Z} \text{ OR } m \text{ is a multiple of } U \\ 0 & \text{else} \end{cases}$$

- Is upsampling

- linear? **yes**
- time-invariant? **no**
- causal? ~~yes~~
- stable? **yes**



$$\max_m |y[m]| = \max_n |x[n]|$$



$$f[n-1] \neq f[m-U] \neq f[n-U] * f[n]$$

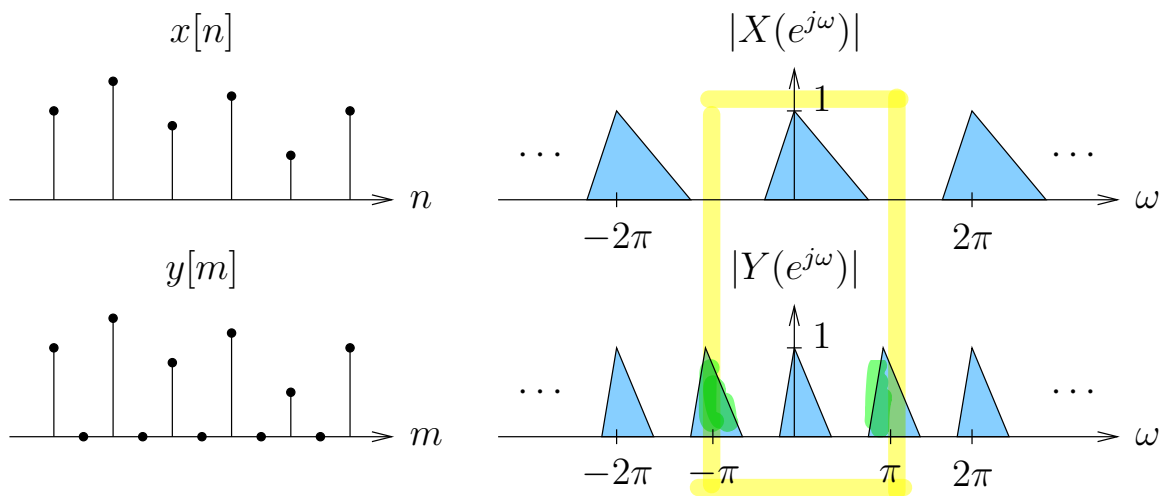
Upsampling (cont.):

In the frequency domain, upsampling can be written as

$$\begin{aligned}
 Y(e^{j\omega}) &= \sum_{m=-\infty}^{\infty} y[m]e^{-j\omega m} \\
 &= \sum_{m: \frac{m}{U} \in \mathbb{Z}} x\left[\frac{m}{U}\right]e^{-j\omega m} \quad (\text{e.g., } m = \dots, 0, U, 2U, \dots) \\
 &= \sum_{n=-\infty}^{\infty} x[n]e^{-j\omega U n} \\
 &= X(e^{j\omega U})
 \end{aligned}$$

$X(e^{j\omega}) \rightarrow X(e^{j\omega U})$
 $U > 1$
 compression

For example, when $U = 2$,

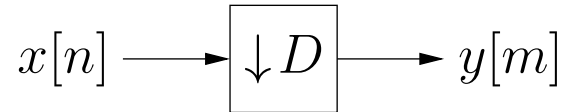


The spectrum is simply compressed by the integer factor U .

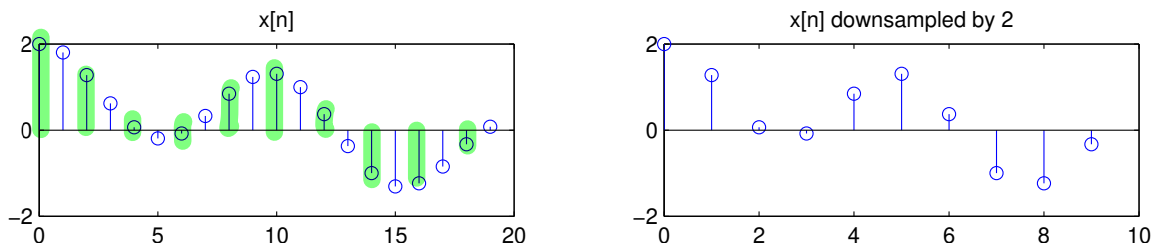
$$\frac{2\pi}{U} - \text{periodic}$$

Downsampling:

- We represent downsampling by the block diagram



- To downsample $x[n]$ by $D \in \mathbb{Z}^+$, we keep every D^{th} sample of $x[n]$ and discard the others.



- In the time domain, downsampling can be written as

$$y[m] = x[Dm]$$

- Is downsampling
 - linear? **yes**
 - time-invariant? **no**
 - causal? ~~yes~~ **no**
 - stable? **yes**

$$\max_m |y[m]| \leq \max_n |x[n]|$$

Downsampling (cont.):

discrete-time
impulse train

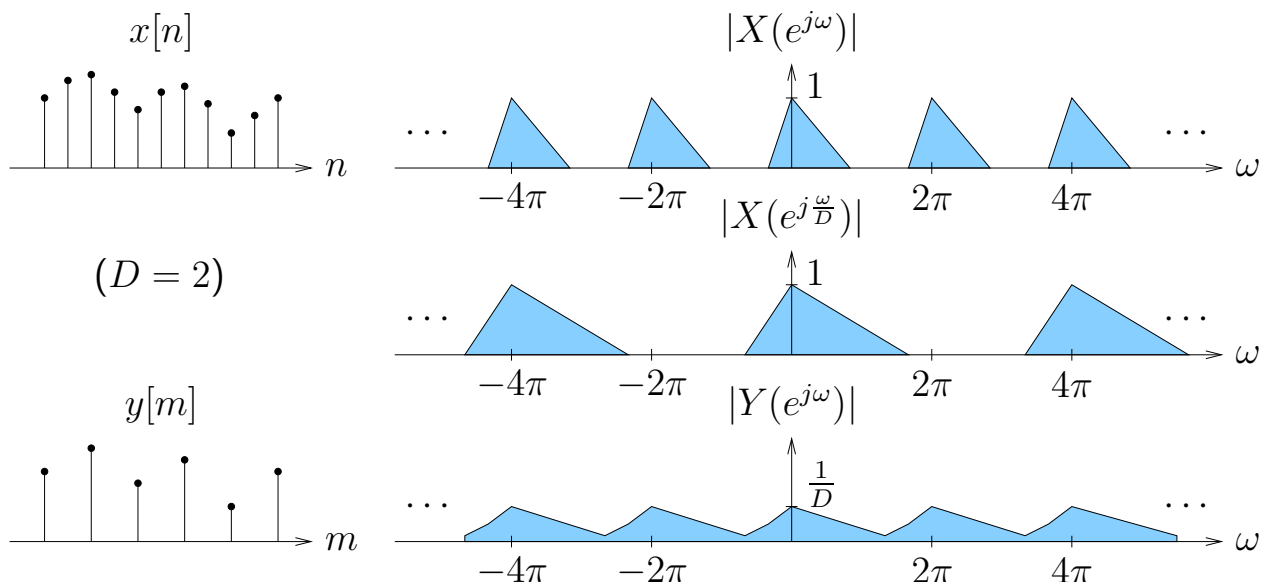
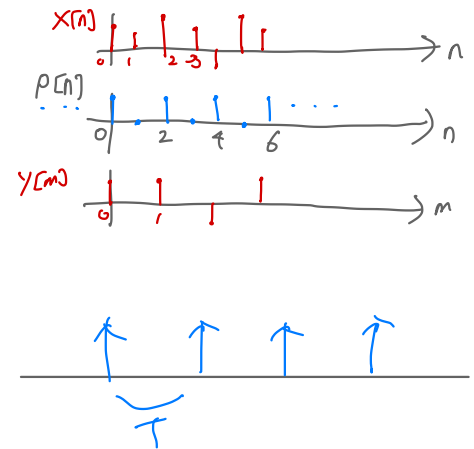
In the frequency domain, downsampling can be written as

$$Y(e^{j\omega}) = \sum_{m=-\infty}^{\infty} \underbrace{x[mD]}_{y[m]} e^{-j\omega m} = \sum_{n=-\infty}^{\infty} x[n] \underbrace{\delta[\langle n \rangle_D]}_{=1 \text{ iff } n \text{ is a mult of } D} e^{-j\omega \frac{n}{D}}$$

$$= \sum_{n=-\infty}^{\infty} x[n] \left(\frac{1}{D} \sum_{k=0}^{D-1} e^{j\frac{2\pi}{D}kn} \right) e^{-j\omega \frac{n}{D}} \quad (\text{next page!})$$

$$= \frac{1}{D} \sum_{k=0}^{D-1} \sum_{n=-\infty}^{\infty} x[n] e^{-j(\frac{\omega - 2\pi k}{D})n}$$

$$= \frac{1}{D} \sum_{k=0}^{D-1} X(e^{j\frac{\omega - 2\pi k}{D}})$$



Spectrum is expanded by D , scaled by $\frac{1}{D}$, and made periodic.